

Elliptic Curve Point Counts over Finite Fields

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Abstract

This project studies elliptic curves over finite fields from the perspective of quadratic residues and character sums. Starting from the elementary number theory concept of the Legendre symbol, we express the number of points on an elliptic curve $E : y^2 = x^3 + ax + b$ over $\mathbb{F}_p$ as a quadratic character sum. We then interpret the Frobenius trace $a_p = p + 1 - |E(\mathbb{F}_p)|$ as the negative of this character sum. Computationally, we compare two methods for computing point counts: direct enumeration of points and the Legendre symbol formula. We also compute normalized Frobenius traces for several elliptic curves and examine their behavior as the prime p varies. Hasse's bound is used not as the main object of verification, but as a theoretical framework that explains why these traces remain within a controlled range. This project connects elementary modular arithmetic with more advanced ideas in computational number theory and elliptic curves.

Keywords: Elliptic curves; Legendre symbol; Character sums; Frobenius trace; Finite fields

1 Introduction

Elliptic curves are important objects in modern number theory. Although they are defined by relatively simple cubic equations, they have deep arithmetic structures and appear in areas such as cryptography, modular forms, and arithmetic geometry [1; 2].

This project begins with ideas from elementary number theory, especially modular arithmetic and quadratic residues. In elementary number theory, the Legendre symbol is used to determine whether a number is a quadratic residue modulo a prime [3]. In this project, we extend this idea to elliptic curves over finite fields. Instead of only asking whether a single congruence has a solution, we study how quadratic residues determine the number of points on an elliptic curve modulo a prime.

The main goal of this project is to connect point counting on elliptic curves with quadratic character sums. For an elliptic curve over $\mathbb{F}_p$, the number of points can be computed directly by checking all possible pairs (x, y) . However, a deeper interpretation comes from observing that, for each value of x , the number of possible values of y depends on whether $x^3 + ax + b$ is a quadratic residue modulo p .

This leads to a character sum formula for the point count and gives a new interpretation of the Frobenius trace. Rather than treating Hasse's bound as the main object to be verified, we use it as a theoretical framework for understanding why these character sums are bounded. Computationally, we compare direct point counting with the Legendre symbol method and then study the behavior of normalized Frobenius traces for several elliptic curves.

Through this approach, the project connects elementary concepts such as quadratic residues and Legendre symbols with more advanced topics such as elliptic curves, character sums, Frobenius traces, and computational number theory.

2 Mathematical Background

In this section, we introduce the mathematical ideas used in this project. The main goal is to show how point counting on elliptic curves over finite fields can be understood through quadratic residues and Legendre symbols.

2.1 Quadratic Residues and the Legendre Symbol

Let p be an odd prime. An integer a is called a quadratic residue modulo p if there exists an integer y such that

$$y^2 \equiv a \pmod{p}. \quad (1)$$

If no such y exists, then a is called a quadratic non-residue modulo p .

The Legendre symbol is defined by

$$\left(\frac{a}{p}\right) = \begin{cases} 1, & \text{if } a \not\equiv 0 \pmod{p} \text{ is a quadratic residue,} \\ -1, & \text{if } a \text{ is a quadratic non-residue modulo } p, \\ 0, & \text{if } a \equiv 0 \pmod{p}. \end{cases} \quad (2)$$

The Legendre symbol allows us to count the number of solutions to the congruence

$$y^2 \equiv a \pmod{p}. \quad (3)$$

Indeed, the number of solutions is

$$1 + \left(\frac{a}{p}\right). \quad (4)$$

If a is a nonzero quadratic residue, then this gives 2 solutions. If a is a non-residue, it gives 0 solutions. If $a \equiv 0 \pmod{p}$, it gives 1 solution.

2.2 Elliptic Curves over Finite Fields

An elliptic curve can be written in the form

$$E : y^2 = x^3 + ax + b, \quad (5)$$

where a and b are constants. To ensure that the curve is nonsingular, we require

$$4a^3 + 27b^2 \neq 0. \quad (6)$$

Over the finite field $\mathbb{F}_p$, the elliptic curve equation becomes the congruence

$$y^2 \equiv x^3 + ax + b \pmod{p}. \quad (7)$$

The number of points on E over $\mathbb{F}_p$, including the point at infinity, is denoted by

$$|E(\mathbb{F}_p)|. \quad (8)$$

2.3 Point Counts as Character Sums

For each fixed $x \in \mathbb{F}_p$, the number of $y \in \mathbb{F}_p$ satisfying

$$y^2 \equiv x^3 + ax + b \pmod{p} \quad (9)$$

is determined by whether $x^3 + ax + b$ is a quadratic residue modulo p . Using the Legendre symbol, this number is

$$1 + \left(\frac{x^3 + ax + b}{p}\right). \quad (10)$$

Therefore, the total number of finite points is

$$\sum_{x=0}^{p-1} \left[1 + \left(\frac{x^3 + ax + b}{p}\right) \right]. \quad (11)$$

After adding the point at infinity, we obtain the character sum formula

$$|E(\mathbb{F}_p)| = p + 1 + \sum_{x=0}^{p-1} \left(\frac{x^3 + ax + b}{p}\right). \quad (12)$$

This formula is important because it shows that point counting on elliptic curves can be understood through quadratic character sums. Thus, the problem is not only about counting points, but also about studying the cancellation behavior of a character sum.

2.4 Frobenius Trace and Hasse's Bound

The Frobenius trace is defined by

$$a_p = p + 1 - |E(\mathbb{F}_p)|. \quad (13)$$

Using the character sum formula, we can rewrite this as

$$a_p = - \sum_{x=0}^{p-1} \left(\frac{x^3 + ax + b}{p} \right). \quad (14)$$

Thus, the Frobenius trace measures the amount of cancellation in the quadratic character sum. Hasse's bound states that

$$|a_p| \leq 2\sqrt{p}. \quad (15)$$

Equivalently,

$$-1 \leq \frac{a_p}{2\sqrt{p}} \leq 1. \quad (16)$$

In this project, Hasse's bound is not treated as the main result to be verified. Instead, it provides a theoretical framework for understanding why the Frobenius traces remain within a controlled range.

3 Computational Method

In this section, we describe the computational methods used in the project. We compute point counts on elliptic curves in two different ways: direct enumeration and the Legendre symbol character sum formula. Comparing these two methods helps connect the computational results with the underlying number-theoretic structure [4].

3.1 Direct Point Counting

The first method is direct enumeration. For a given prime p and elliptic curve

$$E : y^2 = x^3 + ax + b, \quad (17)$$

we check all pairs $(x, y) \in \mathbb{F}_p^2$ and count the pairs satisfying

$$y^2 \equiv x^3 + ax + b \pmod{p}. \quad (18)$$

After counting all finite solutions, we add one point corresponding to the point at infinity. Thus,

$$|E(\mathbb{F}_p)| = 1 + \#\{(x, y) \in \mathbb{F}_p^2 : y^2 \equiv x^3 + ax + b \pmod{p}\}. \quad (19)$$

This method is simple and direct, but it checks p^2 pairs, so its computational cost grows quickly as p increases.

3.2 Legendre Symbol Method

The second method uses the Legendre symbol. For each $x \in \mathbb{F}_p$, we compute

$$\left(\frac{x^3 + ax + b}{p} \right). \quad (20)$$

The point count is then obtained from the formula

$$|E(\mathbb{F}_p)| = p + 1 + \sum_{x=0}^{p-1} \left(\frac{x^3 + ax + b}{p} \right). \quad (21)$$

This method checks only p values of x , rather than all p^2 pairs (x, y) . More importantly, it shows that elliptic curve point counting is closely related to quadratic residues and character sums.

3.3 Frobenius Trace Computation

After computing $|E(\mathbb{F}_p)|$, we calculate the Frobenius trace:

$$a_p = p + 1 - |E(\mathbb{F}_p)|. \quad (22)$$

Equivalently, using the character sum formula,

$$a_p = - \sum_{x=0}^{p-1} \left(\frac{x^3 + ax + b}{p} \right). \quad (23)$$

We also compute the normalized trace

$$\frac{a_p}{2\sqrt{p}}. \quad (24)$$

This normalized value allows us to compare trace values for different primes on the same scale.

3.4 Curves Used in the Experiment

We study three elliptic curves:

$$E_1 : y^2 = x^3 + 1, \quad (25)$$

$$E_2 : y^2 = x^3 - x, \quad (26)$$

and

$$E_3 : y^2 = x^3 + x + 1. \quad (27)$$

These curves are chosen to represent different levels of algebraic structure. The first two curves have more visible symmetry, while the third curve is a more general example. Comparing them allows us to investigate how the coefficients of the curve influence the behavior of Frobenius traces.

3.5 Range of Computation

In this project, we perform computations for prime numbers satisfying

$$p < 500. \quad (28)$$

This range is large enough to observe patterns in the Frobenius traces while keeping the computations manageable.

4 Experiments and Results

In this section, we present the computational results. The experiments have three main goals. First, we compare direct point counting with the Legendre symbol method. Second, we compute Frobenius traces for several elliptic curves. Third, we study the normalized traces to observe how they are distributed within the range predicted by Hasse's bound.

4.1 Comparison of Two Counting Methods

For each elliptic curve and each prime p , we compute the point count in two ways. The first method directly counts all pairs $(x, y) \in \mathbb{F}_p^2$. The second method uses the character sum formula

$$|E(\mathbb{F}_p)| = p + 1 + \sum_{x=0}^{p-1} \left(\frac{x^3 + ax + b}{p} \right). \quad (29)$$

In all computed cases, the two methods give the same point count. This confirms computationally that the Legendre symbol formula correctly captures the number of points on the elliptic curve. More importantly, it shows that point counting is closely related to the behavior of quadratic residues modulo p . Table 1 provides

Table 1: Agreement between Direct Enumeration and Legendre Symbol Method

Curve	p	Direct Count	Legendre Count	Agreement
E1	5	6	6	Yes
E1	7	12	12	Yes
E1	11	12	12	Yes
E1	13	12	12	Yes
E1	17	18	18	Yes

sample computations comparing the direct enumeration method with the Legendre symbol method. In each listed case, the two methods give the same point count. This agreement supports the correctness of the character sum formula used in the computation. It also shows that the Legendre symbol method captures the same point-counting information as direct enumeration, but in a more number-theoretic form. Table 2 compares the computational

Table 2: Runtime Comparison between Direct Enumeration and Legendre Symbol Method

Curve	Prime Range	Number of Primes	Direct Time (s)	Legendre Time (s)	Speedup
E1	$p < 50$	13	0.000841	0.000169	4.98
E2	$p < 50$	14	0.000686	0.000129	5.32
E3	$p < 50$	13	0.000645	0.000105	6.12
E1	$p < 100$	23	0.004195	0.000403	10.41
E2	$p < 100$	24	0.004082	0.000402	10.16

efficiency of the two methods. Although both methods produce the same point counts, their running times are different. Direct enumeration checks all pairs $(x, y) \in \mathbb{F}_p^2$, while the Legendre symbol method checks only the p possible values of x . Therefore, the Legendre symbol method becomes increasingly more efficient as the prime range grows. The speedup values in the table illustrate this computational advantage.

4.2 Frobenius Trace Results

After computing the point counts, we calculate the Frobenius trace

$$a_p = p + 1 - |E(\mathbb{F}_p)|. \quad (30)$$

Figure 1 shows the values of a_p for the three elliptic curves. The dashed curves represent the bounds $\pm 2\sqrt{p}$.

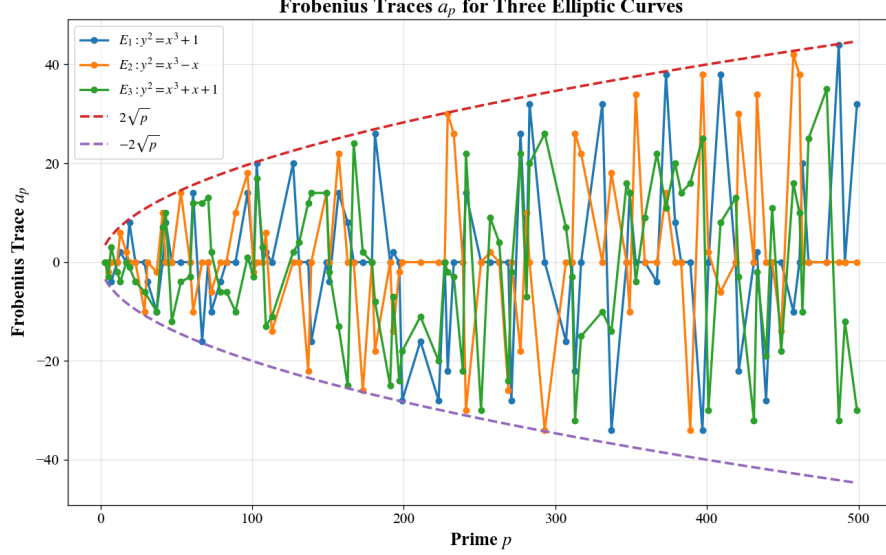


Figure 1: Frobenius traces a_p for the three elliptic curves. The dashed curves represent Hasse's bound $\pm 2\sqrt{p}$.

The figure shows that the Frobenius traces fluctuate around zero. These fluctuations reflect cancellation in the quadratic character sums

$$-\sum_{x=0}^{p-1} \left(\frac{x^3 + ax + b}{p} \right). \quad (31)$$

4.3 Normalized Frobenius Traces

To compare trace values across different primes, we compute the normalized trace

$$\frac{a_p}{2\sqrt{p}}. \quad (32)$$

Figure 2 shows the normalized Frobenius traces for the three elliptic curves. Hasse's bound gives a theoretical limit on the size of these fluctuations. Thus, it illustrates two points at the same time: the traces are bounded by Hasse's inequality, and they fluctuate rather than following a simple monotonic trend as p increases.

By Hasse's bound, all normalized values must lie between -1 and 1 . The figure confirms this behavior for the computed primes. However, the more interesting observation is that the values do not appear constant or uniform in a simple way. Instead, they fluctuate in a way that suggests deeper arithmetic structure. In a broader theoretical context, such normalized Frobenius traces are related to the Sato-Tate distribution [5].

4.4 Comparison of the Three Curves

The three elliptic curves show different trace patterns. The curves

$$E_1 : y^2 = x^3 + 1 \quad (33)$$

and

$$E_2 : y^2 = x^3 - x \quad (34)$$

have more visible algebraic symmetry, while

$$E_3 : y^2 = x^3 + x + 1 \quad (35)$$

is a more general example. Figure 3 shows that these histograms are not to prove a general distribution theorem. Instead, they help us compare the selected curves experimentally. The three histograms show that the normalized

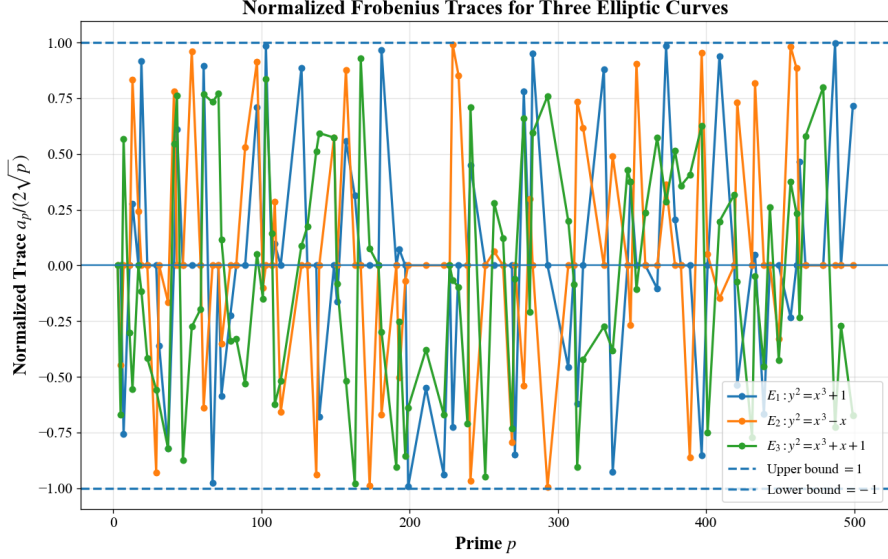


Figure 2: Normalized Frobenius traces $a_p/(2\sqrt{p})$ for the three elliptic curves.

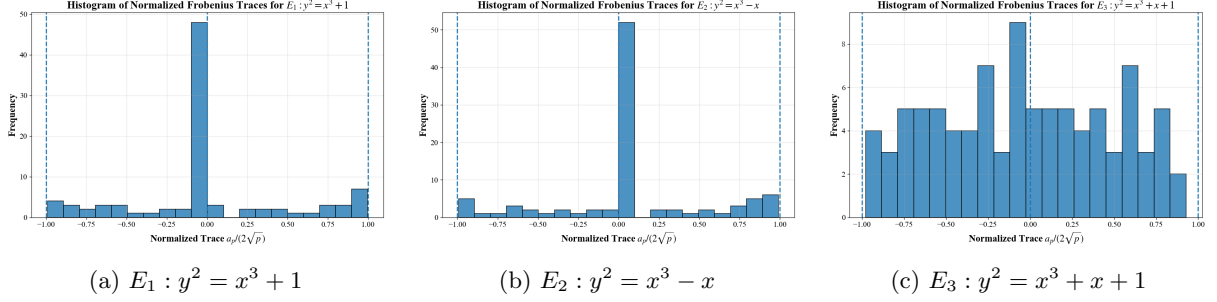


Figure 3: Histograms of normalized Frobenius traces for the three elliptic curves.

traces are distributed differently for the three curves. This suggests that the algebraic form of the elliptic curve may influence not only the size of the Frobenius trace, but also the way these traces are distributed as p varies. In this finite computational range, no simple monotonic pattern is visible, and the distribution appears to depend on the selected curve.

Table 3: Summary Statistics of Normalized Frobenius Traces

Curve	Count	Mean	Std	Min	Max
E_1	93	0.022134	0.483764	-0.992434	0.996915
E_2	94	0.046482	0.470342	-0.993151	0.991228
E_3	93	-0.044282	0.514522	-0.979076	0.928588

Table 3 summarizes the normalized Frobenius traces for the three elliptic curves. The mean values are close to zero for all three curves, which suggests that positive and negative traces tend to balance each other in the computed range. The standard deviations show that the normalized traces are not concentrated at a single value, but spread across a substantial part of the interval $[-1, 1]$.

Although all three curves satisfy the same Hasse bound, their Frobenius traces behave differently. This suggests that the arithmetic behavior of an elliptic curve depends not only on the size of the finite field, but also on the algebraic structure of the curve itself.

This comparison is one of the main motivations for studying normalized Frobenius traces. Hasse's bound describes the possible range of values, but the distribution of values inside that range contains more refined arithmetic information.

5 Conclusion

In this project, we studied elliptic curves over finite fields through the lens of quadratic residues and character sums. Instead of only using direct point counting, we expressed the number of points on an elliptic curve as a sum involving Legendre symbols:

$$|E(\mathbb{F}_p)| = p + 1 + \sum_{x=0}^{p-1} \left(\frac{x^3 + ax + b}{p} \right). \quad (36)$$

This formula connects elliptic curve point counting with elementary number theory. It shows that the Frobenius trace can be interpreted as

$$a_p = - \sum_{x=0}^{p-1} \left(\frac{x^3 + ax + b}{p} \right), \quad (37)$$

which measures cancellation in a quadratic character sum.

Computationally, we compared direct point counting with the Legendre symbol method and found that both methods give the same point counts. We then computed Frobenius traces and normalized traces for three elliptic curves. The results show that the traces fluctuate in a controlled way, remaining within the range described by Hasse's bound.

The purpose of the computation is not simply to verify Hasse's bound, which is already a proven theorem. Rather, the computation helps us visualize how quadratic residues influence point counts and how Frobenius traces behave as the prime varies. This project therefore connects elementary modular arithmetic with more advanced topics in computational number theory and elliptic curves.

A possible extension of this project would be to compute traces for larger primes and compare the distribution of normalized traces with deeper theoretical predictions, such as the Sato-Tate distribution.

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